



AI Engineer 2026 Study Plan

A practical 12-week roadmap for Python, LLM APIs, prompt systems, RAG, LangGraph agents, coding assistants, local inference, UI prototyping, and LLMOps.

12-week path

Wk	Focus	Build proof
1	Python and Git	Build a CLI helper, call APIs, and keep experiments in Git.
2	Math for builders	Vectors, probability, stats, cosine similarity, and evaluation vocabulary.
3	LLM APIs	Compare OpenAI, Claude, Gemini, Qwen, Kimi, and open model APIs.
4	Prompt systems	Task, context, constraints, schema, examples, critique, and stop rules.
5	RAG baseline	Parse, chunk, embed, retrieve, answer with citations, and test retrieval.
6	LangGraph agents	Planner, search, writer, critic, retries, state, and approvals.
7	Advanced agents	CrewAI, AutoGen, OpenHands, OpenClaw, Hermes Agent, and Agents SDK.
8	Coding agents	Ship a tested feature with Claude Code, Codex, Kimi Code, Cursor, or Cline.
9	Vibe-coded UI	Prototype with Lovable, v0, Bolt, Figma Make, Replit, or Gamma.
10	Local inference	Run Ollama and LM Studio; serve with vLLM and log throughput.
11	Evals and monitoring	Add traces, regression prompts, RAG metrics, cost logs, and safety tests.
12	Portfolio capstone	Publish an AI app with README, screenshots, evals, deploy notes, and pitch deck.

Best tools by use case

Use case	Recommended tools
Coding	Claude Code, OpenAI Codex, Kimi Code, Cursor, Windsurf, Cline, Roo Code
Agents	LangGraph, OpenAI Agents SDK, CrewAI, AutoGen, OpenHands, OpenClaw, Hermes Agent
RAG	LlamaIndex, LangChain, Haystack, Chroma, Qdrant, Milvus, Weaviate, rerankers
Local inference	Ollama, LM Studio, llama.cpp, vLLM, Hugging Face models
UI/product	Lovable, v0, Bolt, Figma Make, Streamlit, Chainlit, Next.js, Gamma
LLMOps	LangSmith, Phoenix, MLflow, W&B; Weave, DeepEval, Ragas, Evidently

Prompt enhancement checklist

- Separate task, context, constraints, output format, examples, and quality bar.
- Ask for assumptions and missing information before final output when the task is ambiguous.
- Use a critique pass for weak spots, missing tests, risky assumptions, and better alternatives.
- For coding agents, include files to inspect, commands to run, repo rules, and definition of done.
- For autonomous agents, define tool permissions, stop conditions, budget, memory policy, and escalation rules.

Portfolio project ladder

- Personal document search with citations and retrieval metrics.
- PDF-to-structured-data extractor with validation and failure examples.
- LangGraph research agent with planner, search, writer, critic, and approval checkpoint.
- Local inference lab with Ollama model switcher and vLLM benchmark report.
- Coding-agent feature shipped through Claude Code, Codex, or Kimi Code with tests.
- Portfolio AI product with polished UI, monitoring, eval report, deploy notes, and Gamma pitch.

Recommended learning resources

Resource	Link
AI Python	https://www.deeplearning.ai/short-courses/ai-python-for-beginners/
AI Agents in LangGraph	https://www.deeplearning.ai/short-courses/ai-agents-in-langgraph/
Multi AI Agent Systems with CrewAI	https://www.deeplearning.ai/short-courses/multi-ai-agent-systems-with-crewai/
Building and Evaluating Data Agents	https://www.deeplearning.ai/courses/building-and-evaluating-data-agents/
Claude Code	https://docs.anthropic.com/en/docs/claude-code/overview
OpenAI Codex	https://openai.com/codex/
Kimi Code	https://www.kimi.com/code/en
Ollama	https://docs.ollama.com/
vLLM	https://docs.vllm.ai/en/latest/
Lovable	https://lovable.dev/
Gamma	https://gamma.app/
LangSmith	https://docs.smith.langchain.com/

Definition of ready

- Build a Python or JavaScript AI app from scratch.
- Explain embeddings, attention, tokens, context windows, tool calling, and RAG.
- Use an AI coding agent responsibly and review its diffs.
- Build a LangGraph or equivalent agent with state, tools, and evaluation.
- Run a local model with Ollama and explain when vLLM is useful.
- Deploy a small app and monitor latency, cost, quality, and safety.

Companion files

Interactive HTML: docs/ai-engineer-2026-study-plan.html
Repository README: [README.md](#)